Shivanshu Aryan

New Delhi • cs1240237@iitd.ac.in • +91-9572276455
• shivanshu-aryan • zerodecay357

Education and Academic Details

Indian Institute of Technology, Delhi July 2024 – May 2028
B.Tech in Computer Science and Engineering
◦ CGPA: 7.78/10
◦ **Relevant Coursework:** Computer Architecture, Discrete Mathematics, Data Structures & Algorithms, Digital Logic & System Design, Probability Theory & Stochastic Processes, Calculus, Linear Algebra & Differential Equations.

Scholastic Accomplishments

◦ **JEE Advanced 2024:** Secured AIR 6893 (CRL).
◦ **Indian National Astronomy Olympiad (INAO):** Selected (Top 1.5% from 16,869 participants).
◦ **Indian National Chemistry Olympiad (INChO):** Selected (Top 1.2% from 44,363 participants).
◦ **QRT Quant Challenge 2026:** Selected in **top 12 teams** (Sharpe ratio **0.97**); advanced to 2-month quantitative trading bootcamp out of all IIT Delhi participants.

Research Work

◦ **Strategic Splitting under Hamilton's Apportionment Method** COL8184 · Prof. Rohit Vaish
- Modeled state-splitting as a sequential game under Hamilton's method; derived the Fragment Reduction Lemma and Monotonicity Theorem; characterized equilibrium strategies via backward induction.
- Measuring manipulation frequency — fraction of Dirichlet-sampled apportionment instances where a state gains seats at sequential game equilibrium — across parameters (n, H, α).
- Running simulations on real-world election data (Hong Kong, Nigeria) to validate theoretical predictions.

Projects

◦ **Out-of-Order RISC-V Processor Simulator** COL216 · Computer Architecture
- Implemented full Tomasulo's Algorithm in C++ with ROB, RAT, reservation stations, pipelined execution units (Adder, Multiplier, Divider, Logic, Branch), and Load/Store Queue.
- Supports 2-bit saturating branch prediction, precise exception handling (overflow, divide-by-zero, OOB memory), in-order commit with speculative execution and pipeline flush.

◦ **Hexabot V2, Hexapod RL Locomotion** Robotics Club, IIT Delhi · 2025–26
- Simulating an 18-DOF hexapod in PyBullet with analytical IK, cubic Bézier gait trajectories, and a Gymnasium environment; training PPO (Stable-Baselines3) with curriculum learning.
- Implementing domain randomization (friction, mass, servo latency) for sim-to-real transfer onto Feetech SC-15 serial bus servos via ESP32.

◦ **LevelDB Storage Engine Extensions** COP290 · Prof. Smruti Ranjan Sarangi
- Extended Google's LevelDB with **Range Scan**, **Range Delete** (tombstone-consistent with SSTable semantics), and **Manual Full Compaction** with statistics reporting.

◦ **The Kanban Board** COP290 · Prof. Smruti Ranjan Sarangi
- Full-stack project management app (TypeScript, React + Express + PostgreSQL/Prisma) with JWT auth, role-based access, WebSocket notifications, configurable DB-stored workflow graphs, and server-side WIP enforcement.

◦ **Road Fighter (FPGA Arcade Game)** COL215 · Digital Logic & System Design
- Arcade racing game on Basys3 in Verilog; VGA controller (640×480), 8-bit LFSR for random rival spawns, FSM-based collision and game-over logic.

Technical Skills

Languages:	C++, Python, TypeScript, Verilog/SystemVerilog, MATLAB, Bash, LaTeX, HTML, CSS
Libraries/Frameworks:	React, Express.js, Prisma, NumPy, Pandas, Matplotlib, PyBullet, Stable-Baselines3, TensorFlow (basic)
Tools & Platforms:	Git, CMake, Qt, PostgreSQL, Vivado/Quartus, Arduino/ESP32, Blender, CoppeliaSim, Fusion 360, ProVerif, SymbiYosys
Operating Systems:	Linux (Ubuntu, Arch), Windows

Positions of Responsibility and Extra-Curricular Activities

◦ **Robotics Winner, Inter-Hostel Tech GC** 2024–25
– Achieved **1st place** in Robotics representing hostel in IIT Delhi's technical championship.
– Led team to simulate a 7-DOF robotic arm on CoppeliaSim for the winning solution.

◦ **Design Representative, Alumni Relations Cell, IIT Delhi** June 2025 – Present
– Created visual content (Figma, Canva, Photoshop) for alumni events and outreach.
– Collaborated with alumni team to enhance engagement via strategic design.

Last updated April 2026